ABHISHEK GUPTA

abhishek.gupta6706@gmail.com ⋄ 9569558748 ⋄ LinkedIn ⋄ GitHub

PROFESSIONAL SUMMARY

AI Engineer with experience in designing and developing Generative AI and machine learning solutions for enterprise applications. Skilled in building LLM-powered applications, Retrieval-Augmented Generation (RAG) systems, RESTful APIs with FastAPI, and data-driven AI solutions using Python. Strong foundation in exploratory data analysis (EDA), machine learning algorithms, CNNs, parameter-efficient model design, and transfer learning, complemented by hands-on experience in deploying scalable AI systems. Passionate about translating cutting-edge AI research into practical, high-impact solutions that drive business value.

EDUCATION

Indian Institute of Information Technology, Bhopal	M.Tech in Artificial Intelligence	9.29	2024–2026
Rajkiya Engineering College, Sonebhadra	B.Tech in Electronics Engineering	7.88	2020–2024

EXPERIENCE

Software Engineer *Kellton Tech Solutions* — *May 2026 – Present*

Project: AI-Powered Observability Platform — *June 2026 – Present*

- Developed a pipeline to convert unstructured application logs into a structured format, enabling efficient processing and analysis.
- Generated vector embeddings from the structured log data and stored them in a vector database to support semantic search and retrieval.
- Implemented a hybrid retrieval mechanism that combines semantic vector search with keyword-based retrieval to identify the most relevant log entries based on the user's query.
- Integrated the retrieved context with a Large Language Model (LLM) to generate accurate, human-readable explanations of the root cause, helping engineers quickly understand and resolve system failures.

Project: AI-Powered Cost Estimation Tool — *May 2026 – June 2026*

- Performed EDA to derive insights and align features with cost prediction objectives.
- Performed feature engineering and encoded categorical variables using Label Encoding, followed by training a XgBoost Regressor with hyperparameter tuning using Optuna.
- Achieved strong predictive performance with R^2 scores above 0.75 across all 4 cost output variables.

Trainee *Kellton Tech Solutions* — *Oct 2025 – Apr 2026*

Project: AI-Powered Survey Analytics Platform — *Nov 2025 – Mar 2026*

- Designed and deployed a production-grade, real-time analytics pipeline with MongoDB integration and continuous learning, enabling dynamic model updates on incoming survey data.
- Performed regression and correlation analysis across 10+ question types, along with advanced visualizations (heatmaps, pulse-based analysis) to uncover actionable insights.
- Integrated NLP-based sentiment analysis and Generative AI to generate automated, user-friendly explanations of analytical results for non-technical stakeholders.
- Built REST APIs using FastAPI for all analytics services within the platform, covering regression, correlation, heatmap, sentiment, and video analytics modules.

PROJECTS

Alzheimer's Disease Prediction using Lightweight CNN [Repo Link] — *July 2025 – Oct 2025*

- Designed a lightweight 3-layer CNN for four-class Alzheimer's disease classification using MRI brain images.
- Trained and evaluated models on both ADNI and Mendeley datasets, achieving 96.88% and 98.12% accuracy.
- Conducted bidirectional cross-dataset validation between both datasets, achieving **98% accuracy** in both ways.

SKILLS

- **Programming & Backend:** Python, SQL, FastAPI
- **Machine Learning:** Regression, Classification, Feature Engineering, Model Evaluation, Hyperparameter Tuning
- **Deep Learning:** Neural Networks, CNN, Parameter-Efficient Model Design, Transfer Learning
- **Natural Language Processing:** Text Preprocessing, Sentiment Analysis
- **Generative AI:** Prompt Engineering, LLM-based Insight Generation, Explainable AI
- **Libraries:** NumPy, Pandas, Scikit-learn, TensorFlow, OpenCV, Hugging Face, LangChain, spaCy

- **Tools & Platforms:** Git, GitHub, Linux, Microsoft Azure

ACHIEVEMENT

- Selected among 30,000 applicants nationwide for **Amazon ML Summer School 2025**

RESEARCH WORK

- Lightweight CNN Architecture for Efficient Alzheimer's Disease Classification
- Secure Hardware IP Design Framework for Ownership Protection of IP Vendors and SoC Integrators

CERTIFICATIONS

- **Generative AI Certification** - *Oracle* [Certificate Link] *Sep 2025*
- **AI Foundations Associate** - *Oracle* [Certificate Link] *Sep 2025*
- **Data Science using Python** - *Prutor @ IIT Kanpur* [Certificate Link] *Oct 2022*